

=====

Equipment ID: MRPL-BOIL-501-HP

Equipment Description: Atmospheric Crude Fired Heater & Radiant Steam Coil

Operating Unit: CDU-1 Crude Charge Furnace Section

Inspection Date: 2026-09-09

Thermographer: K. R. Nagesh (Level III Certified Infrared Thermographer)

THERMAL IMAGING & TUBE INTEGRITY FINDINGS:

- Radiant Tube Skin Thermocouple (TI-501-14, South Wall): 722 deg C (Design Limit: 650 deg C).
- Adjacent Skin Thermocouple (TI-501-15): 708 deg C.
- IR Camera Radiometric Hotspot Band: Localized thermal banding 1.8 meters in length on Coil #4.
- Flame Impingement Observation: Burner #6 tile degraded causing direct flame impingement on tubes.
- **Draft Pressure at Arch: -1.2 mm H2O (Recommended range: -2.5 to -4.0 mm H2O).**
- **Tube Sagging & Bowing: Visual laser alignment reveals 28 mm horizontal tube sag on Pass 2.**

METALLURGICAL RISK RATING: EXTREME / HIGH PRIORITY

Operating 9% Cr-1 Mo radiant tubes above 700 deg C accelerates creep rupture and severe internal coking, presenting an imminent furnace tube rupture hazard.

URGENT OPERATIONAL RECOMMENDATIONS:

1. Immediately throttle fuel gas to Burner #6 and bias firing load to east wall burners.
2. Increase steam-to-oil decoking ratio by 10% to enhance internal convective heat removal.
3. Adjust stack damper to restore firebox draft to -3.0 mm H2O to eliminate positive arch pressure.
4. Plan pigging / online steam-air decoking of Radiant Pass 2 during upcoming maintenance.